Governed Feature Selection for Credit-Risk Models: Shapley Attribution with Pre-Registered Parsimony under a Decision-Gate Governance Framework

Ravi*

***Abstract*—Machine-learning credit-risk models are increasingly accurate but remain hard to *govern*: the questions of *which* features are genuinely useful, *how parsimonious* a defensible model may be, and *whether* a selection is adequately powered and fair are usually answered by analyst judgement rather than a reproducible, auditable procedure. We present two coupled contributions. The first is a pre-registered feature-selection methodology that separates order-invariant *attribution* (sampled Shapley importance against a shadow-feature null) from a distinct, fixed *selection* rule (the smallest feature set that is both DeLong-indistinguishable from the full model and within a 5% relative-Gini tolerance), followed by always-on isotonic calibration and a regulatory PD floor. The second is a hash-chained decision-gate governance framework (G0–GF) that wraps it with materiality-tiered rigour, binding every statistical bar to a cited source and separating *governance* decisions from *analytics* parameters, which are never re-opened. Across a corpus of fourteen public and reference credit datasets run under one fixed configuration, we show that the same rule yields genuine parsimony at small sample sizes yet near-full feature sets at large ones (a large-N sensitivity effect we characterise), that excluding a protected attribute is necessary but not sufficient to remove disparate impact, and that signal strength—not sample size—sets the achievable discrimination ceiling. We add the Riley *et al.* minimum-sample-size calculation alongside events-per-variable as a reported sufficiency diagnostic and give a retrospective across the corpus.**

***Impact Statement*—Credit decisions affect access to capital for individuals and firms, and supervisors now require model-risk controls (e.g., the U.S. interagency guidance superseding SR 11-7, and the EU AI Act's treatment of creditworthiness assessment as high-risk). This work shows how feature selection—an early, consequential step usually left to discretion—can be made *reproducible and auditable* without inventing economic assumptions: thresholds and pricing remain institution inputs, and governance gates decide process, not analytics. The framework makes a model's parsimony, calibration, fairness, and data-sufficiency claims traceable to recorded runs and cited bars, which should lower the cost of independent validation and reduce the risk of overstated or unfair models reaching deployment.**

***Index Terms*—Credit risk, feature selection, Shapley values, model governance, calibration, fairness, disparate impact, sample size, explainable AI.**

I. Introduction

MACHINE-learning models now dominate credit-risk scoring, yet their *governance* has not kept pace with their accuracy. A modern scorecard is the product of dozens of consequential choices made before any headline metric is reported—which raw columns to admit, how to bin and screen them, which to keep, how small a model is defensible, and whether the data are even adequate to support it. In practice these choices are made by analyst judgement and documented, if at all, after the fact. That is precisely where model risk concentrates, and precisely what supervisors now scrutinise: the U.S. interagency guidance SR 26-2 (superseding SR 11-7) demands effective, independent challenge of model development; the EU AI Act classifies creditworthiness assessment as high-risk; and Basel CRE36 requires a use test and a per-grade calibration backtest. None of these tells a practitioner *how* to make the early modelling steps reproducible and auditable.

We identify three governance gaps in the feature-selection stage. First, **which features are genuinely useful**: greedy, order-dependent screening produces artefactual retention, and Shapley-style attribution—increasingly the default interpretability tool—fairly apportions importance but does not, by itself, select a subset. Second, **how parsimonious a model may defensibly be**: small differences in discrimination are routinely presented as wins or losses with no inferential test, and the "right" number of features is asserted rather than derived. Third, **whether the data are adequate and the model fair**: sample-size justification rarely goes beyond a rule of thumb, and the removal of a protected attribute is often assumed to remove disparate impact. Each gap is an opening for an overstated or unfair model to reach deployment with a clean-looking paper trail.

We address these with two coupled artefacts. The first is a *pre-registered feature-selection methodology* that cleanly separates order-invariant *attribution* from a fixed *selection* rule, tests feature signal against an empirical shadow-feature null, calibrates by construction, and reports both events-per-variable and the Riley minimum sample size as governed sufficiency diagnostics. The second is a *decision-gate governance framework* that wraps the methodology in a materiality-tiered sequence of gates, each closing with an accountable human sign-off written to a tamper-evident hash chain, and—by design—never lets a per-run governance decision re-open a pre-registered analytical parameter. Evaluated as a single fixed configuration across fourteen credit datasets, the pair exposes behaviour that discretion would hide: the same selection rule yields genuine parsimony at small N but near-full feature sets at large N; excluding a protected attribute does not remove disparate impact; and signal strength, not sample size, bounds achievable discrimination.

Contributions.

1) A **pre-registered feature-selection methodology** separating Shapley *attribution* from a fixed *parsimony* selection rule (§III).
2) A **hash-chained, materiality-tiered decision-gate governance framework** binding every bar to a cited source and separating governance from analytics (§IV).
3) An empirical study over fourteen datasets under one fixed configuration, including the large-N parsimony effect, the "necessary-but-not-sufficient" fairness finding, and a Riley-*et al.* sample-size retrospective (§VI).

II. Related Work

Feature importance and selection. Classical selection splits into filter methods (e.g. Information Value, the credit-scoring standard [1]), wrapper methods, and embedded penalisation (LASSO, elastic net). Shapley-based attribution has become the dominant interpretability tool: SHAP [2] gives local, per-prediction attributions, while global measures such as SAGE [3] and the Shapley Population Variable Importance Measure (SPVIM) [4] attribute model-level

*The author is with the Indian Institute of Foreign Trade, New Delhi, India (e-mail: ravi_phdm23@iift.edu).

Manuscript prepared October 25, 2026. This is a working draft for internal review, not a submitted version.

predictiveness. A recurring confusion—which our methodology is built to avoid—is treating these attributions as a *selection* criterion: a Shapley value fairly *apportions* predictiveness but does not identify a minimal sufficient subset, so an importance ranking must be paired with an explicit selection rule. Boruta's shadow-feature device [5] and stability selection [6] address the related problem of distinguishing real signal from noise; our methodology adopts the shadow-null idea as its retention test.

Parsimony and equivalence. Preferring the smallest model statistically consistent with the best one has a long heritage—the one-standard-error rule of CART [7] and its treatment in the statistical-learning literature [8]. Our methodology operationalises this with a paired-AUC equivalence test [9] plus a relative-Gini tolerance, rather than an information criterion, because discrimination is the quantity credit practice cares about [10], [1].

Model-risk governance. Supervisory expectations for model risk—the U.S. interagency guidance SR 26-2 [11] (superseding SR 11-7), Basel CRE36 [12] (use test, per-grade calibration backtest), the EU AI Act's [13] high-risk regime for creditworthiness assessment, and the EBA's Margin-of-Conservatism guidance—specify *what* must be validated, documented, and independently challenged, but not a concrete, reproducible procedure that couples feature selection to signed, tamper-evident approval. Our governance framework fills that gap by turning each analytical stage into a governed gate with a hash-chained decision log. Documentation artefacts such as model cards [14] and datasheets for datasets [15] standardise *what* to report about a model or data source, but they are descriptive records rather than a gated, append-only decision process; our framework is complementary, producing such records as a by-product of signed gate decisions.

Fairness and disparate impact. Anti-discrimination practice distinguishes disparate *treatment* from disparate *impact*, the latter commonly screened with the four-fifths rule (EEOC [16]). A substantial fairness-in-ML literature formalises group-fairness criteria [17]; a consistent empirical lesson, which our runs reproduce, is that removing a protected attribute is necessary but not sufficient, because correlated features reconstruct it.

Sample-size adequacy. Credit-scoring studies rarely report a sample-size justification beyond the events-per-variable rule of thumb [18]. Recent work shows EPV is too coarse [19] and derives signal-aware minimum sample sizes for prediction models [20], [21], with a C-statistic-to-R^2 route for forward use [22]. Our methodology reports both EPV and the Riley minimum-N as governed diagnostics. In sum, each ingredient exists in isolation; our contribution is their coupling into one pre-registered, reproducible, and auditable pipeline, evaluated here as a fixed-config corpus.

III. The Feature-Selection Methodology

The proposed methodology deliberately separates two questions that are routinely conflated in applied feature selection (the full pipeline and its governance gates are shown in Fig. 1):

- **Attribution** — "How much predictive value is attributable to this feature, *order-invariantly*?" Answered by a Shapley importance measure against an empirical null.
- **Selection** — "What is the *smallest* feature set whose performance is statistically indistinguishable from the full set?" Answered by a separate, pre-registered stopping rule.

The separation is principled, not stylistic. Shapley values are a *fair-attribution* solution concept (efficiency, symmetry, dummy, additivity); they do not identify an optimal subset. A redundant-but-informative feature still receives a positive share, and a feature whose value concentrates in rare coalitions can average below any bar despite being needed. Attribution therefore *ranks*; a separately justified rule *selects*. Greedy, category-sequential screening—the common alternative—produces order-dependent, artefactual retention; an order-invariant importance estimate plus an explicit selection rule avoids it.

Algorithm 1 Governed feature selection (one dataset, fixed configuration)

Require: data (X, y); learner; value function v; shadows $m \geq 10$; tolerance τ=0.05; level α=0.05; confirmation threshold c=0.80; seed
screen features by $\mathrm{IV} \geq 0.02$ with monotonic WOE binning ▷ Stage 1
group columns at $|\rho| \geq 0.7$ into coalitional players P ▷ Stage 2
add m shadow players: row-permuted copies of real columns
if $|P| + m \leq 15$ **then**
    compute exact Shapley values ϕ for all players
else
    estimate ϕ by sampled permutations to a target MC s.e.
end if
$b \leftarrow$ 95th percentile of the shadow players' ϕ ▷ empirical null
$C \leftarrow \{\, p : \phi_p > b$ and bootstrap confirm. freq $\geq c \,\}$
rank C by ϕ descending ▷ Stage 3
for $k = 1, \ldots, |C|$ **do**
    fit top-k on out-of-fold folds; record Gini_k and DeLong p_k vs full
end for
$k^\star \leftarrow \min\{\, k : p_k \geq \alpha$ and $(\mathrm{Gini}_{\text{full}} - \mathrm{Gini}_k)/\mathrm{Gini}_{\text{full}} \leq \tau \,\}$
calibrate the top-$k^\star$ model (isotonic); apply PD floor/cap ▷ Stage 4
Ensure: selected set (top-$k^\star$ of C), calibrated model, additive reason codes

Inference target. We estimate population feature importance *with respect to a specified learner and value function*: how much cross-validated predictiveness is recoverable from a feature set, for a given learner class, on a given population—*not* how much a feature "matters to credit risk" in the abstract. Formally, with value function $v(S)$ equal to the cross-validated predictiveness of the columns in S, the importance of feature j is the Shapley value

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[v(S \cup \{j\}) - v(S) \big], \qquad (1)$$

which is the Shapley Population Variable Importance Measure (SPVIM) of Williamson and Feng [4]. This is *not* SAGE [3], which marginalises absent features out of a single fitted model; here each coalition S is *refit*.

The pipeline has five stages (0–4), summarised in Algorithm 1; the governance framework that signs off on their outputs is described in §IV.

A. Stage 0: Eligibility and data governance

Stage 0 runs *before any statistics*: it attaches a per-column classification that travels with every downstream result, and excludes columns that cannot be modelled—leakage, identifiers, constants and junk—each by a *computable precondition* recorded with its measured value (Table I), so every exclusion is reproducible. Protected and proxy attributes, by contrast, are tagged rather than dropped: each is labelled `permissible`, `prohibited` (e.g. sex, age, marital status), `proxy_risk` (e.g. geography, education, household composition), or `unclear`. On these public research benchmarks, protected and proxy attributes are *retained with mandatory disclosure* so results remain comparable with the large published literature on the same datasets; no model here is intended for live use. Stage 0 also records point-in-time availability (flagging `near_target` and `external_score` columns that make a model a behavioural or

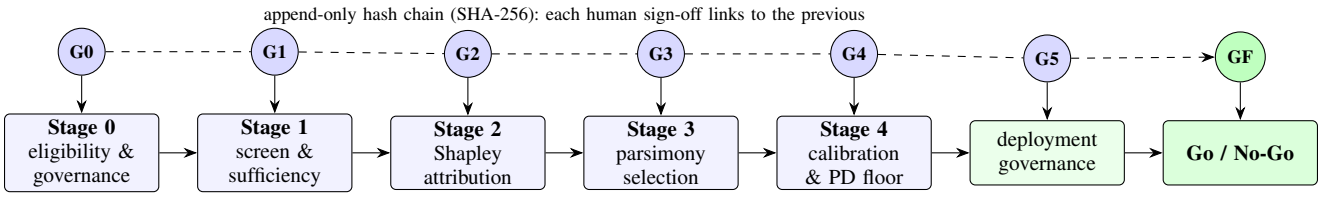


Fig. 1. The governed pipeline. The feature-selection methodology (blue, Stages 0–4) runs under a sequence of governance gates (G0–G5), each closed by an accountable human sign-off that is written to an append-only hash chain; the committee gate GF rules the final Go/No-Go with all gate evidence visible. Gates decide scope, eligibility, fairness treatment, cut-offs and disposition—never the pre-registered analytical parameters (P8).

TABLE I
STAGE-0 ELIGIBILITY EXCLUSIONS. EACH IS DRIVEN BY A COMPUTABLE PRECONDITION; THE SEMANTIC CATEGORIES PAIR THAT SIGNAL WITH A RECORDED RATIONALE, NEVER A SILENT DROP. EXAMPLES FROM THE SBA RUN.

Category	Computable precondition	Example (SBA)
Constant	nunique ≤ 1 (zero variance)	`State`, `BalanceGross`
Identifier / PII	cardinality $\to n$ (a key), or high-cardinality free text	`LoanNr` (2102/2102), `Name`
Leakage	target agreement $\to 1$, or sentinel event-rate lift $\geq 3\times$; confirmed post-outcome by meaning	`MIS_Status` (100%), `ChgOffPrinGr` (99.5%)
Junk	isna().all() (all-blank), or absent from the data dictionary	`xx`

score-recovery exercise rather than an application scorecard), audits sentinel codes and shared missingness for leakage, notes the accept-only sampling that truncates feature distributions [23], and declares *mandatory* columns and a missingness policy that are enforced only at scoring (§III-F). Two further Stage-0 *reports* were added with the staged protocol (§IV-C); neither excludes a column or adds a pass/fail bar. A *blank-versus-outcome scan* gives, for every column with blanks, the share blank, the event rate among blank and among filled rows, and a Fisher exact p-value against a Bonferroni reference. A *look-ahead scan* flags column definitions containing a temporal phrase (e.g. "in 3 years") so that a human reads them against the question of whether they could use information after the statement date, and records a verdict.

B. Stage 1: Univariate screening

Information Value (IV) is the single a-priori screen, at the conventional threshold IV ≥ 0.02 [1]; Cramér's V, χ^2 and p-values are reported descriptively and decide nothing. Binning is monotonic Weight-of-Evidence (WOE) with a minimum bin size of 5% of rows,

$$\mathrm{WOE}_i = \ln \frac{g_i/G}{b_i/B}, \qquad \mathrm{IV} = \sum_i \Big(\tfrac{g_i}{G} - \tfrac{b_i}{B}\Big)\mathrm{WOE}_i, \qquad (2)$$

where g_i, b_i are goods/bads in bin i. Monotonicity is enforced only where it is supported: a genuinely U-shaped feature whose IV recovers $\geq 20\%$ when the constraint is freed is tagged `shape_nonmonotonic` and routed to a single-column model check rather than being judged on its monotonic IV. A reliability gate (χ^2 minimum expected cell ≥ 5) and a bin-count sensitivity check (WOE vs 5/10/20 quantile bins) escalate borderline columns to the same model check, whose verdict is a Gini confidence interval excluding zero.

C. Stage 2: Order-invariant attribution

The default value function is repeated cross-validated AUC and the default learner is a penalised logistic model on monotonic-WOE inputs (`woe_logistic`)—the credit-scoring standard form: deterministic and hyperparameter-insensitive, which removes refit noise from the Shapley estimate. XGBoost [24] is run as a robustness learner; material disagreement in the two rankings (reported via Kendall's τ) is itself a finding about interaction dependence. Class reweighting is *off* by default: on GMSC it inflated the mean predicted probability to 0.286 against a true base rate of 0.067, and the resulting miscalibration is avoided rather than repaired.

Two devices make the attribution both tractable and honest. First, correlated columns are not pruned (which would reintroduce a greedy univariate criterion); instead, hierarchical clustering on $1 - |\rho_{\text{Spearman}}|$ cut at $|\rho| = 0.7$ makes *each cluster a single Shapley player*, preserving the efficiency property over the full candidate set. Second—the central methodological fix—significance testing of Shapley values is replaced by a **shadow-feature null** [5]: at least ten shadow players (row-permuted copies of real columns) are added, the identical Shapley computation is run over the augmented set, and a real player is `importance_confirmed` only if its value exceeds the 95th percentile of the shadow distribution *and* its bootstrap confirmation frequency is ≥ 0.80. The bar is a quantile of an empirical null, so no multiple-testing correction is needed and fold-dependence does not bias it. Benjamini–Hochberg [25] p-values and Cohen's d [26] are retained only as reported diagnostics—usefully self-refuting, since on German Credit a permuted-noise shadow column received $p = 0.0000$ with a *negative* Shapley value, which a significance rule would have confirmed and the shadow bar correctly rejected. Enumeration is exact when the player count (including shadows) is ≤ 15 and sampled otherwise, with the Monte Carlo standard error reported per player.

D. Stage 3: Pre-registered parsimony

The eligible set is the confirmed players, ranked by ϕ_j. For $k = 1, \ldots, K$ the model on the top-k players is scored on *out-of-fold* predictions (so every k is evaluated on the same rows and DeLong's paired test is valid [9]), and the full parsimony curve is a required output. The stopping rule is fixed in advance: choose the smallest k that is both statistically indistinguishable from the full-candidate model by DeLong's test at $\alpha = 0.05$ *and* within a relative-Gini tolerance,

$$k^\star = \min\Big\{k : \; p_{\text{DeLong}}(k) \geq 0.05 \;\wedge\; \tfrac{\text{Gini}_{\text{full}} - \text{Gini}_k}{\text{Gini}_{\text{full}}} \leq 0.05\Big\}. \qquad (3)$$

If no k qualifies, that is reported as the result ("no parsimonious subset meets the tolerance")—it is never replaced by the best available k. Because the two conditions are combined by *and*, the rule behaves very differently across sample sizes (§VI): at small N both clear together and yield genuine parsimony, whereas at large N DeLong detects the loss from dropping even a weak feature and the full set is returned.

E. Stage 4: Calibration, reason codes, and sufficiency diagnostics

The selected model is calibrated by cross-validated isotonic regression [27], [28] (always on), and calibrated probabilities are used whenever a PD is quoted to a person. Because the learner is WOE-logistic, the score is an exact additive decomposition $\text{logit} = \beta_0 + \sum_j \beta_j \, \text{WOE}_j(x)$, giving faithful per-feature *reason codes* with no post-hoc approximation. A regulatory PD floor of 0.03% (Basel CRE32.3) and a 99.9% cap are applied at scoring, as are the Stage-0 mandatory-field gate and missingness policy. Finally, data sufficiency is reported—never used to auto-reject—through two diagnostics side by side: events-per-variable [18], [19] and the Riley *et al.* minimum sample size [21], [20], the latter derived from the anticipated C-statistic and prevalence via the Cox–Snell R^2 approximation of [22]. A shortfall is a disclosed caveat the governance layer weighs, not a gate failure—consistent with the principle that selection parameters are fixed and only governance decides disposition.

IV. The Governance Framework

The methodology of §III produces a defensible *model*; it does not, by itself, produce a defensible *decision to use that model*. A supervisor or internal audit function asks questions the analytics cannot answer: who decided the data were adequate, and could they have decided otherwise? Who chose this feature set, on what recorded rationale? Who is accountable if the model is wrong, and were they independent of the builder? Where is the evidence that each regulatory expectation was *met*, not merely claimed? The governance framework answers these by turning each methodology stage into a *governed decision gate*: the machine emits a recommendation and a full evidence pack, a human with the right role and sufficient independence chooses from a closed set of genuinely viable options, and the choice is written to a tamper-evident log. This mirrors modern model-risk expectations—the U.S. interagency guidance SR 26-2 (superseding SR 11-7), the EU AI Act's treatment of creditworthiness assessment as high-risk, and Basel CRE36—without importing their numeric bars by assumption: every threshold used is tied to a cited source.

A. Design principles

Eight principles govern the framework; three are load-bearing. **(P1) Machine recommends, human decides, record proves**: every gate is recommendation + evidence pack + closed options + accountable sign-off + log entry. **(P2) Segregation of duties**: the Develop, Validate, and Approve functions are never held by one person for one model. **(P8) Gates decide governance, not analytics**: a gate's option set may *never* re-open a pre-registered analytical parameter (estimator mode, selection tolerances, shadow-bar quantile, confirmation frequency, `random_state`, CV folds). Those are fixed in the methodology, identical across every dataset, and chosen once; gates decide scope, eligibility, fairness treatment, cutoffs, and Go/No-Go. P8 is what keeps the method reportable: letting a per-run gate switch the estimator would be the dataset-dependent, path-changing behaviour a research methodology must avoid. The single failure mode the framework is explicitly designed against is *rubber-stamping*—hence the anti-strawman rule below and the independence requirements in §IV-E.

B. The decision gates G0–GF

The methodology's stages become gates G0–G5, with a final committee gate GF (Table II). G0–G5 are *stage gates* (a single accountable role signs each); GF is the *committee gate*, ruled by the Model Approval Committee. Crucially, GF is last and is decided only after G4's per-grade calibration backtest and G5's cutoff policy and acceptance demonstration are complete and visible—one does not approve deployment and then check whether the PDs are honest and the decisions sensible.

TABLE II
The decision gates and the methodology stage each governs.

Gate	Stage	Core question decided
G0	Stage 0	Is the dataset fit to build on? (coverage, label integrity, scope)
G1	Stage 1	Which features survive screening; is N adequate (EPV + Riley, reported)?
G2	Stage 2	Is the attribution stable enough to select on?
G3	Stage 3	Which feature set is adopted—parsimonious vs full, and the trade-off?
G4	Stage 4	Calibrated, discriminating, and fair enough to approve? (per-grade backtest)
G5	Scoring	Deployment controls + cutoff policy + acceptance demonstration
GF	whole model	GO / GO-WITH-CONDITIONS / NO-GO (committee)

A model is assigned a *materiality tier* (HIGH / MEDIUM / LOW) at G0, after SR 26-2's exposure-$\times$-purpose logic; the tier sets *who* signs each gate and *how high the pass-bar sits*, but never changes the analytics—every metric is measured identically across datasets. Lowering a tier is itself a recorded G0 decision.

C. Staged execution: each stage stops at its gate

The gates of Table II could be walked after the whole pipeline has run, which is how the fourteen corpus runs were executed (§V). That order has a cost: if an early gate changes direction, the later stages have already been computed, and the signer has seen downstream numbers before deciding upstream gates. The staged protocol removes both problems. Each methodology stage runs, then *stops at its gate*; the next stage starts only after the accountable role has chosen an option, given a rationale of at least twenty characters, and the entry has been appended to the hash chain. A gate cannot be sealed before its stage has run, and a stage refuses to run while the previous gate is unsealed. A gate that changes direction at G0–G3 voids its own entry and every later one (kept in the audit trail) and re-runs from the stage it governs; a change at G4, G5 or GF needs no re-run of the methodology. The staged run calls the same primitives with the same arguments and seed as the one-shot pipeline, and an all-approved staged run reproduces the one-shot candidates, selected columns, parsimony curve, shadow bar, calibration and backtest (an automated equivalence test). Consequently P8 is unaffected: staging changes *when* a human decides, never *what* is computed.

Three further changes move an assessment earlier rather than adding a bar. Events-per-variable and the Riley minimum N are reported on every basis reached (candidate columns, confirmed players, selected model) at G1, G2 and G3. The regulatory acceptance checklist (rows R1–R10) is derived as evidence appears, so the final gate reads a completed checklist; derived statuses are proposals that the signer may override with a recorded note. And the holders of the signing roles are declared at G0, so that one person holding several roles is visible when it occurs and not only at GF. At G5 the acceptance pack is accompanied by a coverage report of which scoring controls the data actually exercises.

D. Anatomy of a gate

Each gate produces one Gate Decision Record with five parts: (i) the machine recommendation and its numeric basis; (ii) the

TABLE III
EXAMPLE GATE DECISION RECORD—THE CLOSED, VIABLE OPTION SET AT G3 (SELECTION) FOR CREDITRISK, EACH WITH THE RESIDUAL RISK THE SIGNER ACCEPTS. THE MACHINE MUST DEFEND EVERY OPTION; UNVIABLE "STRAWMAN" OPTIONS ARE NOT PRESENTED.

Option	Key data	Residual risk accepted
A (chosen): ratify k=6 benchmark	Gini 0.76; DeLong p=0.88; rel. loss -0.01%	benchmark carries circular features; not origination-valid
B: origination-only 4-feature model	Gini 0.61 (-19%); independent of incumbent	lower discrimination for application-time use
C: NO-GO	—	delay; a solid independent signal exists, so unwarranted

full evidence pack (the actual numbers, traceable to a run, not a summary); (iii) a closed set of 2–4 options, each carrying its own data, rationale, benefits, risks, and an explicit *residual-risk acceptance*; (iv) a human sign-off (role, identity, exactly one option, free-text rationale, timestamp); and (v) an append-only log entry. The *anti-strawman rule* forbids padding: if the machine cannot construct a genuine case for an alternative, it presents fewer options and flags "single viable option; escalate," routing the decision upward rather than manufacturing fake choices.

Tamper-evidence is a hash chain over the decision records. With a genesis token $h_0 = \mathsf{GEN}\text{-}\langle\text{TAG}\rangle\mathsf{0}$, each gate's entry hash is

$$h_i = \text{SHA-256}\big(h_{i-1} \,\|\, c_i\big)[0{:}8], \tag{4}$$

where c_i is the gate's decision content and $\|$ is concatenation. Any later edit to a sealed decision breaks every subsequent link, so a reversal must be a *new* recorded decision, never a silent edit—the audit trail is append-only. Table III shows a representative decision record: the closed, viable option set at the selection gate for CREDITRISK, each option carrying the residual risk the signer explicitly accepts.

E. Roles, segregation of duties, and the committee

The framework defines seven roles; the hard rule (P2) is that Develop, Validate, and Approve are never the same person for one model, mirroring SR 26-2's conflict-of-interest provisions. Validators sign G1–G4 and can *return* a gate; **G4 requires dual sign-off** (Risk Officer *and* Validator), because calibration and fairness are where model error becomes capital error and conduct risk. The Model Approval Committee—chaired by the CRO or an independent delegate, with a quorum that is void without an independent validator present—rules GF. Any member who developed the model recuses and is minuted; the developer presents but does not vote. Every MAC decision is itself a hash-chained event.

F. The Go/No-Go engine

At GF a deterministic engine synthesises six inputs—dataset adequacy, the signed gate decisions, residual risk vs appetite, the regulatory checklist, audit expectations, and the new-model acceptance bar—into a recommendation the committee rules on: *all gates signed and every regulatory criterion satisfied and residual risk within appetite* ⇒ **GO**; *remediable partials with conditions* ⇒ **GO-WITH-CONDITIONS** (the engine lists the conditions); *any gap on a must-have criterion, or an unsigned/returned gate, or residual risk outside appetite* ⇒ **NO-GO** (the engine names the blocking item and the gate to return to). The engine never upgrades a NO-GO; the committee may downgrade on judgement and must minute why, but cannot clear a hard gap except by resolving it as a new, re-synthesised gate decision.

V. EXPERIMENTAL SETUP

Datasets. The corpus is fourteen binary-outcome credit datasets (Table IV): widely used public benchmarks—including the UCI German [29], [30] and Taiwan [31] credit datasets, the Home Equity (HMEQ) [32] and Give-Me-Some-Credit [33] sets, the FICO HELOC panel [34], a LendingClub extract [35], the PAKDD-2009 Brazilian retail-credit set [36], and the US Small Business Administration (SBA) loan set [37]—alongside additional reference portfolios, the Polish companies bankruptcy set [38] (the corpus uses the file whose outcome is checked five years ahead; UCI numbers it "Year 1") and a US mortgage performance panel [32] (the latter reshaped from 622,489 loan-period rows to one origination-time row per loan). They span $N \in [1{,}000,\ 149{,}391]$, prevalence $[4\%, 52\%]$, and candidate pools of 9–87 features. Per Stage 0, protected and proxy attributes are retained with disclosure so results remain comparable with the published literature on these datasets; no model here is intended for live use. Full provenance and target definitions are recorded per dataset.

Fixed configuration (identical for every dataset, by P8). Primary learner `woe_logistic` (XGBoost as a robustness learner); value function cross-validated AUC; correlated columns grouped at $|\rho| = 0.7$; ≥ 10 shadow players with a 95th-percentile bar and confirmation frequency ≥ 0.80; exact Shapley when players (including shadows) ≤ 15, sampled otherwise; selection rule (3); isotonic calibration; PD floor 0.03% and cap 99.9%. The random seed is 42 throughout. No gate re-opens any of these.

Evaluation protocol. Splits are stratified $80/20$ taken *before* any resampling, with the test set held at its natural base rate; the parsimony curve is scored on out-of-fold predictions from stratified 5-fold cross-validation so DeLong's paired test is valid. Discrimination is reported as Gini (primary) with AUC, KS, partial AUC, Brier and ECE [39]; paired model comparisons use DeLong [9]; calibration uses Hosmer–Lemeshow [40] and a per-grade realised-vs-expected backtest; fairness uses the adverse-impact ratio against the four-fifths rule; data sufficiency uses EPV [18] and the Riley minimum-N [21].

Governance and reproducibility. Each corpus dataset passes through gates G0–GF with a human sign-off recorded at every gate and appended to the hash chain (4); the materiality tier is set at G0. The fourteen corpus runs were executed *one-shot*: every methodology stage was run and the gates were then walked in order. The staged protocol (§IV-C) was adopted afterwards and is used for the Polish horizon series reported separately in §VI-F; the corpus runs were not re-executed under it. The pipeline is deterministic: independent re-runs of two datasets reproduced their committed results exactly—German (Gini 0.54, GO-WITH-CONDITIONS) including its hash chain, and CREDITRISK (selection Gini 0.76), the latter re-run on a *cleaned* copy of the file (its corrupt label column removed) yet returning the identical screen, attribution, and k=6 selection, confirming that the result depends only on the modelling inputs. Validation of the corpus runs is i.i.d. cross-validation only—no corpus run is evaluated out of time, so no claim of out-of-time or production readiness is made—and all accept/decline thresholds and pricing are treated as exogenous institution inputs, never derived here.

VI. RESULTS

We run fourteen public and reference credit datasets through the full governed exercise (G0–GF) under *one fixed configuration* (seed 42; selection = smallest k with DeLong $p \geq 0.05$ and $\leq 5\%$ relative-Gini loss; sampled Shapley when players > 15). The corpus spans N

TABLE IV
THE GOVERNED CORPUS. POOL = FEATURES ENTERING THE PIPELINE AFTER STAGE-0; SEL = SELECTED; GINI/AUC ARE OUT-OF-FOLD AT SELECTION. VERDICT IS THE SEALED GF DECISION.

Dataset	N	base	pool→sel	Gini	AUC	Verdict
SBA	2,102	.326	19→11	0.86	0.93	GO-cond
HMEQ	5,960	.200	12→11	0.81	0.91	GO-cond
POLISH-5y	7,027	.039	64→37	0.78	0.89	GO-cond
CREDITRISK	32,581	.218	9→6	0.76	0.88	GO-cond
GMSC	149,391	.067	9→8	0.70	0.85	GO-cond
HELOC	10,459	.522	23→17	0.59	0.79	GO-cond
TAIWAN	30,000	.221	20→14	0.54	0.77	GO-cond
GERMAN	1,000	.300	17→5	0.54	0.77	GO-cond
MORTGAGE	50,049	.303	10→6	0.46	0.73	GO-cond
MSME	1,707	.060	87→15	0.43	0.72	**NO-GO**
LC	39,717	.142	21→9	0.36	0.68	GO-cond
LOAN	9,578	.160	13→7	0.34	0.67	GO-cond
TH02	1,225	.264	14→6	0.27	0.64	GO-cond
PAK	50,000	.261	39→6	0.15	0.57	**NO-GO**

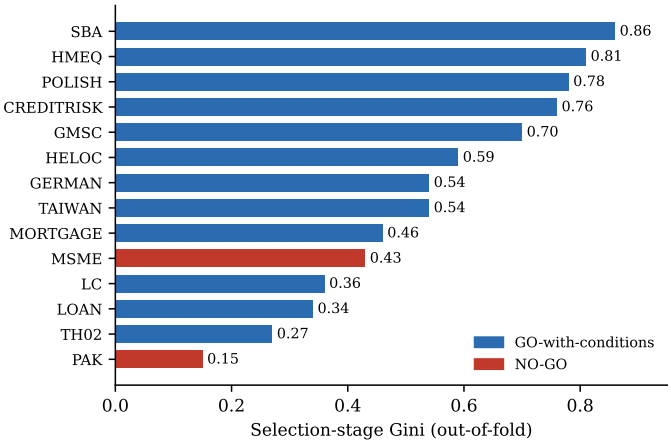


Fig. 2. Selection-stage Gini across the governed corpus, coloured by sealed GF verdict. Discrimination spans an order of magnitude; the two NO-GO runs are not simply the lowest-Gini—MSME is rejected on sufficiency, not discrimination.

from 1,000 to 149,391, prevalence from 4% to 52%, and candidate pools from 9 to 87 features. Table IV and Fig. 2 summarise it; twelve runs seal GO-WITH-CONDITIONS and two (MSME, PAK) seal NO-GO. Every figure traces to a recorded run. The Polish set appears once, as the five-year-ahead file (POLISH-5y); the one-year-ahead file is run separately under the staged protocol (§VI-F).

A. A worked run: German credit, G0 to GF

To make the pipeline concrete, we trace one dataset end-to-end. On German credit (N=1,000, 30% default rate), **G0** excludes the three protected attributes (age, sex/marital status, foreign-worker status) and sets a MEDIUM materiality tier, leaving a 17-feature pool. **G1** accepts the univariate screen—12 of 17 features pass, and both sufficiency diagnostics are met (EPV ≈17.6; Riley minimum 323 vs 1,000 available)—with `checking_status` dominant (IV 0.67). **G2** confirms all 12 players above the shadow-feature null; the U-shaped `credit_amount`, which a monotonic IV screen under-rates, is vindicated at the model check and enters the selected set. **G3** selects k=5 features (`checking_status`, `credit_history`, `months_duration`, `savings_status`, `credit_amount`), Gini 0.54—the smallest k at which the equivalence test (p=0.11) and the 5% tolerance (4.6%) clear *together*. **G4** finds calibration sound (Hosmer–Lemeshow p=0.11, ECE 0.03) and fairness mixed: the sex adverse-impact ratio is clean (0.93) while the age ratio fails four-fifths (0.34) *despite age being excluded*—a true-risk-driven disparity reconstructed through correlated features. **G5** fixes the scoring controls (mandatory `checking_status`; exact reason codes; PD floor), and **GF** rules GO-WITH-CONDITIONS. Every decision is signed and appended to the hash chain, which seals at a final digest; an independent re-run reproduced every gate and the chain exactly (§V).

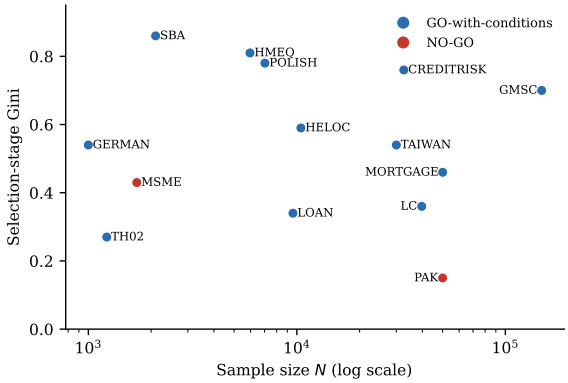


Fig. 3. Signal, not volume, sets the ceiling: each dataset's selection-stage Gini against its sample size (log scale), coloured by sealed verdict. More rows do not raise the achievable discrimination—German (1,000 rows) outscores LC (40,000) and GMSC (149,391).

B. Parsimony varies with sample size

The single fixed stopping rule (3) produces strikingly different parsimony depending on N, because DeLong's power to detect the loss from dropping a weak feature grows with sample size. At **small** N the rule genuinely parsimonises: on German (N=1,000) it selects 5 of 17 features, with both conditions clearing together (DeLong p=0.107 ≥ 0.05 and relative-Gini loss 4.56% ≤ 5%). At **large** N the same rule returns a near-full set: on LC (N≈40,000) it keeps 8 of 9 players (DeLong p=1.0, loss 0.0%), and on HMEQ (N≈6,000) DeLong rejects *every* proper subset (k=9 vs full, p=6.1×10^{-4}) even though a 5%-loss criterion alone would admit a six-feature model—so the *and*-rule returns all eleven. This is the methodology declining to overstate parsimony when the data can resolve the difference. We call this a *large-sample significance effect*: the same rule, opposite behaviour at the two ends of the sample-size range, both honest. A corollary visible in Table IV is that *signal, not volume, sets the ceiling* (Fig. 3)—German reaches Gini 0.54 on 1,000 rows while LC reaches 0.36 on 40,000.

C. Fairness: excluding a protected attribute is necessary but not sufficient

Fairness is identified by a fixed *proxy-leakage* procedure: we partition the held-out set by a protected attribute the model *does not use*, take the model's favourable-decision rate per group on the internal master scale (indicative—no institution cutoff), and report the adverse-impact ratio against the four-fifths rule; where a group fails, we test whether the model is accurate *within* each group, which separates true-risk-driven disparity from proxy bias. On two independent datasets, removing the protected attribute did not remove disparate impact, because correlated features reconstruct it. On **LC**, geography (`addr_state`) was dropped at a cost of only 0.007 Gini, yet a residual regional approve-rate disparity persisted (worst adverse-impact ratio 0.62, below the four-fifths bar). On **German**, `age` was excluded at G0, yet the age-band approve-rate AIR was 0.34 (under-25 default rate 41.3% vs 25.5% for 60+); because the model is

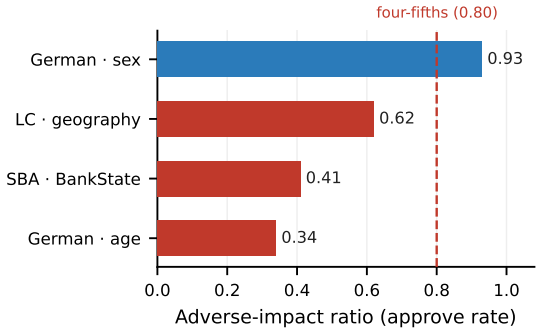


Fig. 4. Adverse-impact ratio after the protected attribute is excluded, against the four-fifths bar (0.80). Sex is clean; age (German) and geography (LC, SBA) fail despite the attribute being excluded—disparate impact reconstructed through correlated features, true-risk-driven in each case.

accurate *within* each band, this is disparate *impact driven by true risk*, not proxy bias. Both runs therefore record the same governed finding (Fig. 4)—proxy removal is necessary but not sufficient, and a fairness-constrained replication or reject-inference study would be required before any live use—surfaced as a binding GF condition rather than discovered post-deployment.

D. Calibration

With class reweighting off and isotonic calibration always on, calibrated probabilities are tight across the corpus: expected calibration error is 0.020 on HMEQ, 0.012 on LC, 0.033 on German and 0.007 on MORTGAGE, and Hosmer–Lemeshow is *not* rejected even at the largest N (GMSC 149,391, LC). The one exception is the origination-only MORTGAGE model (N=50,049), where HL *is* rejected at g=10 ($p\approx 0$) but passes at g=5 (p=0.34) with the per-grade backtest clean (8/8 grades inside the Wilson interval): the large-N hypersensitivity that dominates parsimony surfaces for calibration only on the coarsest- fitting model, and there the per-grade realised-vs-expected backtest—not HL—is the governing test (as recorded at G4 for that run). *Which backtest.* The per-grade backtests sealed for the fourteen corpus runs were computed on out-of-fold predictions. A stricter variant, which fits the isotonic calibrator on one part of the data and tests the *deployed* calibrated model on a held-out part, was introduced later (first with the mortgage hazard framing) and is used in §VI-F; the earlier runs have not yet been re-verified under it, so the two are not directly comparable. Because the learner is WOE-logistic, each PD carries an exact additive reason-code decomposition. Fig. 5 illustrates the per-grade reliability on CREDITRISK, whose model spans the full AAA–D range: observed default rates track the predicted probabilities closely across nearly two orders of magnitude.

E. Sample-size adequacy (EPV and Riley)

Table V reports the two sufficiency diagnostics. Thirteen of fourteen runs meet the Riley minimum; the lone shortfall is **MSME** (requires 3,654, has 1,707), which is also the only run failing *both* diagnostics and the only HIGH-tier NO-GO—so the single failure is the case already rejected. The two diagnostics diverge in both directions, which is exactly Riley's value over EPV. Below EPV 10 sit *two* runs—MSME (6.8) and **POLISH-5y** (7.3)—yet only MSME also fails Riley: POLISH's strong signal (Gini 0.78) clears the shrinkage criterion despite thin events (271), so it meets Riley with room to spare. The converse is **PAK**, which has an enormous EPV (2,173) yet needs 4,041 rows because its weak signal (AUC

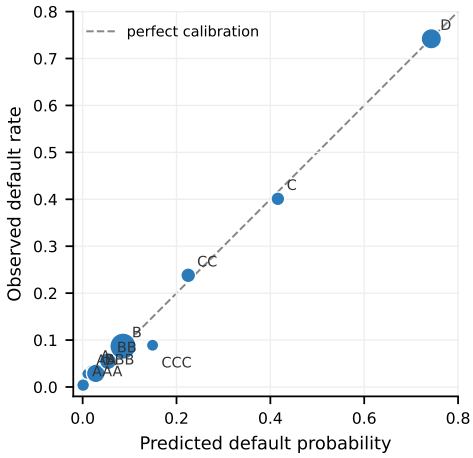


Fig. 5. Per-grade calibration reliability on the CREDITRISK held-out set (rating grades AAA–D; marker area $\propto$ grade population). Points lie close to the diagonal across the full risk range; Hosmer–Lemeshow is not rejected (p=0.44, ECE 0.011). The one mild outlier is the small CCC grade, which over-predicts (a conservative direction).

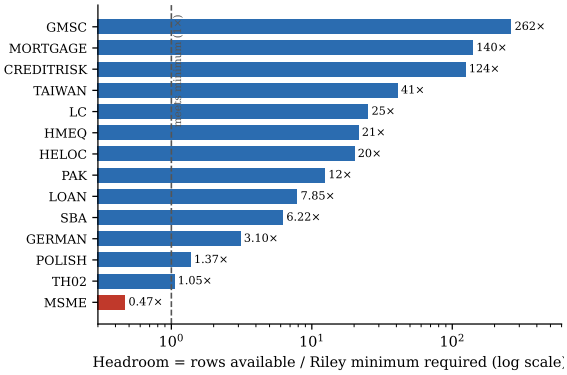


Fig. 6. Sample-size headroom: rows available divided by the Riley minimum required (log scale; dashed line is the 1× meets-the-minimum point). Only MSME falls short (0.47×); PAK clears by 12× despite a huge EPV, because its weak signal raises the requirement.

0.57 $\Rightarrow$ R^2_{cs}=0.013) makes shrinkage demand volume; and **TH02** passes by only +55 rows, a thinness EPV (54) does not convey. Fig. 6 shows the headroom as rows available over rows required.

F. A staged run: Polish bankruptcy, one year ahead

The Polish file whose outcome is checked one year ahead (5,910 firms, 410 bankruptcies, 6.94%; 64 financial ratios; seed 42, same fixed configuration) was the first dataset run under the staged protocol, with every gate signed before the next stage ran. It is not part of Table IV: its firms and event rate differ from the five-year-ahead file, so the two are not a paired comparison, and no test of whether discrimination falls with horizon has been made.

Screening passes 63 of 64 ratios (the dominant ratio, sales growth, has information value 2.19). Attribution confirms 19 players above the shadow bar (0.00161; weakest 1.74×). Selection chooses k=4 players spanning 25 columns: Gini 0.826 (95% CI 0.797–0.857) against 0.838 for the full set, DeLong p=0.255, relative loss 1.4%. EPV is 6.4 on the candidate pool and 16.4 on the selected model, which meets the Riley minimum with a 5.2× margin. The deployed-model backtest passes 9 of 10 grades on 1,182 held-out firms; grade B under-predicts risk (16.7% realised against 8.3% expected), and Hosmer–Lemeshow does not reject with ten groups (p=0.072)

TABLE V
RILEY MINIMUM-N RETROSPECTIVE (REPORTED DIAGNOSTIC). $N_{\text{REQ}} = \max(B1, B3, B4)$.

Dataset	N	N_{req}	EPV	Meets?
SBA	2,102	338	62	yes
HMEQ	5,960	278	108	yes
POLISH-5y	7,027	5,141	7.3	yes (+1,886)†
CREDITRISK	32,581	263	1185	yes
GMSC	149,391	570	1251	yes
HELOC	10,459	522	321	yes
TAIWAN	30,000	737	474	yes
GERMAN	1,000	323	60	yes
MORTGAGE	50,049	358	2526	yes
MSME	1,707	3,654	6.8	**SHORT**
LC	39,717	1,581	625	yes
LOAN	9,578	1,220	219	yes
TH02	1,225	1,170	54	yes (+55)
PAK	50,000	4,041	2173	yes

† POLISH-5y meets the Riley minimum but its EPV $7.3 < 10$ (events per selected column; its record also reports 4.2 on the 64-candidate screening pool)—strong signal clears the shrinkage criterion despite thin events, the mirror image of PAK. MSME alone fails both diagnostics.

but rejects with five (p=0.019). The sealed verdict is GO-WITH-CONDITIONS.

The run also shows what the staged reports are for. The ratio 'sales growth' is blank for 103 firms, 99 of them bankrupt (96.1%; Fisher $p < 10^{-100}$), so a model that scores a blank as a value assigns those firms near-certain default. The pattern is a data-collection artefact of unknown cause, not a financial ratio, and it was noticed only at the deployment gate because the blank-versus-outcome scan did not yet exist; with the scan it appears at G0. The deployed control refers a blank application instead of scoring it. On complete rows the in-sample AUC is 0.895 against 0.920 on all rows (descriptive, not out-of-fold), so part of the reported discrimination rests on the pattern. Two further items stay open: one profitability ratio is defined over 'gross profit in 3 years', and the source does not say whether those years follow the statement date; and bankrupt and operating firms were sampled in different calendar windows. The acceptance pack exercised the PD floor but never the 99.9% cap (the highest PD among scored firms is well below it), and one real bankrupt firm is rated in the best grade.

VII. DISCUSSION

Governance without contaminating the analytics. The central design choice—that gates decide scope, eligibility, fairness treatment and disposition but never the analytical method (P8)—is what makes the corpus comparable at all. Because the estimator, tolerances, shadow bar and seed are fixed across every dataset, a difference in outcome is a property of the data, not of a per-run tuning decision. The governance layer buys traceability and accountability; it does not buy, and must not buy, the freedom to change the method until the answer improves.

What the findings mean for practice. This large-sample effect is not a defect of the rule but a faithful consequence of it: at large N a paired-AUC test can resolve the loss from dropping even a weak feature, so an honest equivalence criterion returns a fuller model. Practitioners should therefore expect genuine parsimony mainly at modest sample sizes and treat a large-N "parsimonious" claim with suspicion. The fairness result sharpens a familiar warning: because correlated features reconstruct an excluded attribute, proxy removal is a floor, not a remedy, and the responsible posture is to *measure and disclose* residual disparate impact (here as an indicative adverse-impact ratio, with the accept/decline cutoff left exogenous) and bind a fairness-constrained follow-up, rather than to claim fairness by omission.

Limitations. This study validates a *procedure*, not a performance claim: we do not yet assert that the methodology selects better subsets than established alternatives—that requires the pre-specified benchmark against LASSO, elastic net, RFE, Boruta and TreeSHAP, which is implemented but not run across the corpus. Validation of the corpus runs is i.i.d. cross-validation; none is evaluated out of time, so no out-of-time or production-readiness claim is made. All datasets are accept-only samples, so feature effects are subject to range restriction [23]. The governance roles are exercised by a single researcher in this methodological setting; the framework specifies the *structure* of independent challenge rather than demonstrating a multi-party deployment. In the staged Polish run the researcher chose every option, but the rationale text was drafted by an AI assistant, shown before sealing, and recorded as such; the independence rows of the acceptance checklist (R7, R8) are therefore partial. The Polish run also leaves two data questions open (a possible look-ahead in one ratio and a sampling-window confound between bankrupt and operating firms), and its discrimination partly rests on a blank-value pattern whose cause is unknown (§VI-F). Finally, the Riley diagnostic uses the selected-feature count as its parameter count and a closed-form approximation that can under-state the requirement at very high C; neither affects any verdict here, but both are disclosed.

VIII. FUTURE WORK

Several directions follow directly from the limitations above. *Comparative evaluation.* The pre-specified benchmark against LASSO, elastic net, recursive feature elimination, Boruta and TreeSHAP—implemented but not yet run across the corpus—will test whether the methodology selects *better* subsets than established alternatives, upgrading the present "defensible" claim to a comparative one. *Temporal validation.* An exploratory discrete-time (hazard) analysis of the mortgage panel with an out-of-time split exists outside the reported corpus; extending it, and applying the pipeline to other datasets with a reliable time index, would support out-of-time validation and a survival / competing-risks extension, replacing the current i.i.d. cross-validation scope and supporting a production-readiness claim. *Reject inference.* Because all datasets are accept-only, incorporating reject inference would correct the range restriction that attenuates feature effects, and would sharpen the fairness analysis in particular. *Fairness-constrained selection.* Since excluding a protected attribute does not remove disparate impact, a natural extension searches for a less-discriminatory alternative—through fairness-constrained selection or a penalised objective—and reports the discrimination cost of each constraint, making the fairness trade-off an explicit, governed decision. *Multi-party governance.* Exercising the role separation with genuinely independent signers, integrated with a live model-risk workflow, would let us measure the governance overhead and the framework's effect on validation turnaround. *Sample-size refinement.* For high-discrimination settings where the closed-form Riley bound can under-state the requirement, a simulation-based minimum sample size and a candidate-pool parameter count would tighten the reported sufficiency diagnostic.

IX. CONCLUSION

We have presented a pre-registered feature-selection methodology that separates Shapley attribution from an explicit parsimony rule and reports its own calibration, fairness and sample-size adequacy, wrapped in a hash-chained, materiality-tiered governance framework that records who decided what, on what evidence, and under which cited bar. Run as one fixed configuration across fourteen credit

TABLE VI
THE FIXED CONFIGURATION (IDENTICAL FOR EVERY DATASET; SEED 42).

Parameter	Value
Primary / robustness learner	WOE-logistic / XGBoost
Value function	cross-validated AUC
Correlation grouping	single player at $\lvert\rho\rvert \geq 0.7$
Shadow players	≥ 10; bar = 95th percentile
Confirmation frequency	≥ 0.80
Exact vs sampled Shapley	exact if players ≤ 15, else sampled
Selection rule	smallest k: DeLong $p \geq 0.05$ *and* rel. Gini loss $\leq 5\%$
Calibration	isotonic, 5-fold
PD floor / cap	0.03% / 99.9%
Split	stratified $80/20$ pre-resampling; 5-fold OOF
Random seed	42

datasets, the pair surfaces findings that analyst discretion would obscure—sample-size-dependent parsimony, disparate impact that survives protected-attribute removal, and the primacy of signal over volume—while inventing no economic assumptions: thresholds and pricing remain institution inputs. We hope the reproducible, auditable discipline demonstrated here lowers the cost of independent model validation and raises the bar for what counts as a defensible credit model.

APPENDIX: REPRODUCIBILITY

The configuration is fixed across every dataset and chosen once (Table VI), with the random seed 42 throughout, and no governance gate re-opens any of it (P8). Reproducibility rests on this fixed configuration together with the determinism of the pipeline: independent reruns of two datasets reproduced their committed results exactly (§V). Each gate decision is additionally recorded in an append-only, hash-chained log (§IV, Eq. 4) so that any later edit to a sealed record is detectable; the per-run decision logs accompany the paper's artefact. For staged runs the artefact also holds the chain and the stage outputs of each run, and an automated test checks that the staged runner reproduces the one-shot pipeline.

REFERENCES

[1] N. Siddiqi, *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring*. Wiley, 2006.

[2] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[3] I. Covert, S. M. Lundberg, and S.-I. Lee, "Understanding global feature contributions with additive importance measures (SAGE)," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[4] B. D. Williamson and J. Feng, "Efficient nonparametric statistical inference on population feature importance using Shapley values," in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.

[5] M. B. Kursa and W. R. Rudnicki, "Feature selection with the Boruta package," *Journal of Statistical Software*, vol. 36, no. 11, pp. 1–13, 2010.

[6] N. Meinshausen and P. Bühlmann, "Stability selection," *Journal of the Royal Statistical Society: Series B*, vol. 72, no. 4, pp. 417–473, 2010.

[7] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, *Classification and Regression Trees*. Wadsworth, 1984.

[8] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. Springer, 2009.

[9] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[10] S. Lessmann, B. Baesens, H.-V. Seow, and L. C. Thomas, "Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research," *European Journal of Operational Research*, vol. 247, no. 1, pp. 124–136, 2015.

[11] Federal Reserve, OCC, and FDIC, "Supervisory guidance on model risk management (SR 26-2)," Interagency guidance, U.S. prudential regulators, 2026, supersedes SR 11-7 (2011).

[12] Basel Committee on Banking Supervision, "CRE: Calculation of RWA for credit risk," Bank for International Settlements, Basel Framework.

[13] European Parliament and Council, "Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (AI act)," 2024.

[14] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, "Model cards for model reporting," in *Proc. Conf. on Fairness, Accountability, and Transparency (FAT*)*, 2019, pp. 220–229.

[15] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford, "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.

[16] U.S. Equal Employment Opportunity Commission, "Uniform guidelines on employee selection procedures," 29 C.F.R. Part 1607, 1978.

[17] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

[18] P. Peduzzi, J. Concato, E. Kemper, T. R. Holford, and A. R. Feinstein, "A simulation study of the number of events per variable in logistic regression analysis," *Journal of Clinical Epidemiology*, vol. 49, no. 12, pp. 1373–1379, 1996.

[19] M. van Smeden, K. G. M. Moons, J. A. H. de Groot *et al.*, "Sample size for binary logistic prediction models: beyond events per variable criteria," *Statistical Methods in Medical Research*, vol. 28, no. 8, pp. 2455–2474, 2019.

[20] R. D. Riley, K. I. E. Snell, J. Ensor *et al.*, "Minimum sample size for developing a multivariable prediction model: Part ii — binary and time-to-event outcomes," *Statistics in Medicine*, vol. 38, no. 7, pp. 1276–1296, 2019.

[21] R. D. Riley, J. Ensor, K. I. E. Snell *et al.*, "Calculating the sample size required for developing a clinical prediction model," *BMJ*, vol. 368, p. m441, 2020.

[22] R. D. Riley, B. Van Calster, and G. S. Collins, "A note on estimating the Cox-Snell R^2 from a reported C statistic (AUROC) to inform sample size calculations for developing a prediction model with a binary outcome," *Statistics in Medicine*, vol. 40, no. 4, pp. 859–864, 2021.

[23] D. J. Hand and W. E. Henley, "Statistical classification methods in consumer credit scoring: a review," *Journal of the Royal Statistical Society: Series A*, vol. 160, no. 3, pp. 523–541, 1997.

[24] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.

[25] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.

[26] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Lawrence Erlbaum Associates, 1988.

[27] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proc. 8th ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining (KDD)*, 2002, pp. 694–699.

[28] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proc. 22nd Int. Conf. on Machine Learning (ICML)*, 2005, pp. 625–632.

[29] U. Grömping, "South German Credit Data: Correcting a widely used data set," Beuth University of Applied Sciences Berlin, Reports in Mathematics, Physics and Chemistry 4/2019, 2019, uCI Machine Learning Repository dataset #522 (CC BY 4.0).

[30] D. Dua and C. Graff, "UCI machine learning repository," University of California, Irvine, School of Information and Computer Sciences, 2019, http://archive.ics.uci.edu/ml.

[31] I.-C. Yeh and C.-h. Lien, "The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients," *Expert Systems with Applications*, vol. 36, no. 2, pp. 2473–2480, 2009.

[32] B. Baesens, D. Rösch, and H. Scheule, *Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS*. Wiley, 2016.

[33] Kaggle, "Give me some credit," Kaggle competition dataset, 2011, https://www.kaggle.com/c/GiveMeSomeCredit.

[34] FICO, "Explainable machine learning challenge (HELOC dataset)," FICO Community, 2018, https://community.fico.com/s/explainable-machine-learning-challenge.

[35] LendingClub, "LendingClub loan data," Public dataset, accessed via public mirrors.

[36] PAKDD, "PAKDD 2009 data mining competition: credit risk assessment," Pacific-Asia Conf. on Knowledge Discovery and Data Mining, 2009.

[37] M. Li, A. Mickel, and S. Taylor, ""should this loan be approved or denied?": A large dataset with class assignment guidelines," *Journal of Statistics Education*, vol. 26, no. 1, pp. 55–66, 2018.

[38] M. Zięba, S. K. Tomczak, and J. M. Tomczak, "Ensemble boosted trees with synthetic features generation in application to bankruptcy prediction," *Expert Systems with Applications*, vol. 58, pp. 93–101, 2016.

[39] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proc. 29th AAAI Conf. on Artificial Intelligence*, 2015, pp. 2901–2907.

[40] D. W. Hosmer and S. Lemeshow, "Goodness of fit tests for the multiple logistic regression model," *Communications in Statistics — Theory and Methods*, vol. 9, no. 10, pp. 1043–1069, 1980.